GRAY HAIR SCALP SERUM

50 mL / PEPTIDE SERIES / PROPOSED LEAVE-ON FORMULA



New white airless-bottle and carton concept. Proposed presentation; materials, delivery system and label require validation.

One focused candidate

DARKENYL 1.00% + ARCOLYS 1.00% + SILVERFREE 1.50%
With panthenol, betaine and a lightweight aqueous base. Percentages refer to supplied commercial blends by weight.

Development status: proposed formula and packaging for chemist/manufacturer evaluation. Finished-product efficacy, stability, safety and shelf life remain to be established. This revision replaces S01 as the development request; final manufacturing release requires approved specifications.

The product to develop

A complete brief for one serum and one 50 mL presentation. Sensory and performance descriptions below are development targets.

Factory description

Please develop a lightweight leave-on scalp serum for VETUD Peptide Series using candidate formula VTD-RGH-S02. The product should spread through parted hair with controlled application, leave little residue and fit a repeat-use scalp-care routine. Assess the proposed three-active system for compatibility, tolerability and finished-serum performance before recommending scale-up.

Attribute	Development requirement
Product identity	VETUD Peptide Series - Gray Hair Scalp Serum. Working name and artwork pending market-specific review.
Intended development population	Adults with early signs of hair graying. VETUD has not yet been clinically tested in this population.
Texture and appearance	Lightweight, low-viscosity aqueous serum; minimal tack and visible buildup. Final color/clarity depend on qualified active grades.
Fragrance and color	No deliberately added fragrance or colorant requested. Confirm constituents carried by every blend.
Presentation	One 50 mL light-protective pack, preferably an opaque airless pump with narrow scalp nozzle.
Use directions	Dose, frequency, scalp area, drying instructions and storage advice to be established from the final formula, delivery system and safety/tolerability work.
Technical owner	Qualified formulator to lock trade grades, process, specifications and testing; factory to quote and manufacture the approved version.

Intended benefit to investigate

Evaluate whether consistent use changes the appearance or density of gray hair compared with a matched vehicle. Pigment-related claims require evidence for the finished serum and review for each launch market. The development intent is not a proven consumer promise.

Revision approach: retain the candidate active levels while rebuilding the product specification, packaging and validation plan. New appearance and documentation do not establish stronger biological performance.

One formula, by weight

Code VTD-RGH-S02. Nominal percentages for a 1 kg development batch. Confirm exact commercial grades before compounding.

Ingredient	% w/w	1 kg / g	Function / control
Darkenyl, supplied commercial blend	1.00	10.00	Pigmentation candidate; grade and density verification. [1,4]
Arcolys, supplied commercial blend	1.00	10.00	Botanical pigmentation candidate. [5]
SILVERFREE, supplied commercial blend	1.50	15.00	Palmitoyl Dipeptide-52 peptide blend. [6]
Propanediol	4.00	40.00	Humectant / solvent contribution.
Glycerin	2.00	20.00	Humectant; add blend carry-over to total.
Betaine	1.00	10.00	Humectant.
Panthenol, active-content basis	0.50	5.00 active	Adjust weighed amount to purchased grade.
Hydroxyethylcellulose	0.25	2.50	Texture; define grade and hydration method.
Allantoin	0.10	1.00	Conditioning; verify dissolution.
Phenoxyethanol / ethylhexylglycerin 90:10	1.00	10.00	Provisional preservative system. [7]
Purified water, nominal allowance	87.65	876.50	Final water q.s. after all corrections.
Nominal total	100.00	1000.00	Before pH solution / diluted-grade corrections.

Final mass balance: bring to 100.00% with purified water after accounting for pH-adjustment solutions and grade carriers. For example, 0.50% active panthenol requires 0.667% of a 75% grade, so the extra 0.167% replaces water. Reconcile all carriers in the master formula.

Dose interpretation: 1.50% SILVERFREE is not 1.50% pure peptide. Only SILVERFREE among the three named blends supplies the identified peptide. The Darkenyl paper used 1% v/v; do not describe 1% w/w as an exact match without the relevant densities. [1,6]

Preservative balance: 1.00% of the 90:10 blend supplies 0.90% phenoxyethanol and 0.10% ethylhexylglycerin. Calculate any carry-over, check launch-market limits and test protection of the complete formula. No overdose was established by this review.

Resolve the process before the pilot

Development sequence to be finalized by the qualified formulator. This page is not a validated manufacturing instruction.

Required supplier confirmation / pH and temperature

Obtain current limits for each exact active grade before a full-formula pilot. A historical Givaudan Darkenyl leaflet specifies addition below 40 C and at pH below 5. That is an incorporation instruction, not a demonstrated finished-storage pH range. Replace the former default pH 5.0-5.5 with a documented, shared supplier-approved process and final-pH window. No substitute number is authorized here. [4]

1 / Qualify and reconcile the inputs

Record manufacturer, grade, batch, CoA, TDS/SDS, current full INCI, composition ranges, carrier solvents, preservatives and storage requirements. For active blends request available marker/assay specifications and density. Approve traceability and define acceptable substitutions through a documented reformulation review.

2 / Prepare the vehicle

Reserve water for final balance. Select the HEC grade for the applicator and low-residue target; define wetting, hydration, mixing and any permitted base-heating step. Dissolve the conditioning materials using a suitable method. Confirm uniformity, no polymer lumps and no allantoin crystals before active incorporation.

3 / Incorporate within qualified limits

Set the base to the approved addition pH and temperature. Add the commercial blends one at a time according to their current instructions. Record actual temperature, pH and appearance after each addition. Add the preservative at its validated stage. Do not infer a common processing limit from one ingredient alone.

4 / Set the final bulk and fill

Adjust only within the qualified final-pH window; bring to final mass, mix uniformly and deaerate if required. Measure appearance, odor, pH, viscosity, density and yield. Define bulk hold time and filling controls from development evidence. Fill the approved pack and retain labeled samples.

If supplier conditions do not overlap: return a justified revision of the active system and supporting calculations. The chemist should resolve the formulation question before a larger batch is made.

Define success before scale-up

Ask the laboratory to return methods, numerical results, justified acceptance criteria and signed reports. Set limits from suitable standards and actual development data.

Workstream	What to measure	Evidence required to advance
Compatibility screen	Vehicle, single-active diagnostic samples and complete candidate: color, odor, pH, viscosity and precipitation.	Identify adverse changes and establish a workable complete-formula process. Diagnostic samples are not additional products.
Preservation / microbial quality	Preservation efficacy, e.g. ISO 11930 with applicable amendment; routine microbial quality against the current applicable standard. [8]	Qualified-lab reports, methods, organism panel, criteria and results. Preservation efficacy and routine counts are separate requirements.
Stability / package	Real-time and justified accelerated/transport conditions in the intended pack: pH drift, viscosity, crystals, color, odor, leaks and delivery. Active markers where feasible.	Document trend data and a justified storage/shelf-life proposal. Screening alone does not establish 24 months.
Safety / tolerability	Ingredient exposure review, finished-product safety assessment and intended-use scalp tolerability.	Qualified assessor conclusion and a protocol using the intended formula, dose and delivery system.
Performance	Finished serum versus matched vehicle; reproducible gray-hair measurements and standardized photographs.	Prespecified protocol, sufficient sample size, clinically interpretable effect and appropriate analysis of missing data.

Finished-product study design

Plan an independently assessed, randomized and blinded vehicle-controlled study, with approximately 4-6 months as a planning estimate. Define dose, frequency, reproducible scalp sites, gray-hair counts, proximal-shaft pigment assessment, tolerability and adherence. Control hair dye and surface staining. Predefine the primary outcome and analysis before enrollment.

Proposed release specification register

Return justified limits and test methods for appearance, color, odor, pH, viscosity (instrument, settings and temperature), density, microbial quality, fill quantity, package integrity and delivered dose. Approve ranges after pilot measurements; no arbitrary pH or viscosity limits are supplied here.

Claim scope: an advantage over vehicle can support a suitable finished-product claim. A synergy claim requires a component-comparison design; superiority requires a suitable direct comparison.

A product the factory can quote

Preferred concept: opaque white 50 mL airless bottle, narrow scalp nozzle, protective cap and matching carton. Packaging performance remains to be validated.

Front-panel hierarchy

VETUD
PEPTIDE SERIES
GRAY HAIR
SCALP SERUM
50 mL / 1.69 fl. oz.

Supporting carton text

DARKENYL + ARCOLYS + SILVERFREE
PRO-VITAMIN B5
vetud.com



Factory requirement	Specification to return
Bottle and applicator	Material construction, product-contact components, light protection, pump/nozzle type, priming, dose output, residual product, closure and leakage results.
Fill and size	One 50 mL presentation. Fill mass = 50 mL x measured density (g/mL). Confirm net volume, fill tolerances and pack dimensions using the actual component.
Carton and finish	White base, navy type, cobalt/silver accents and flowing hair-strand motif. Quote board, print process, finish and a vendor-specific dieline after component selection.
Final label fields	Market-appropriate full INCI, substantiated directions, warnings, storage, responsible-party details, batch coding and other required fields. Finalize after formula and launch markets are locked.

Artwork status: the image is a visualization, not a print-ready dieline or proof that the depicted pump is commercially available. Confirm trade-name usage, artwork tolerances, legibility and the net-content declaration before printing. Use the approved VETUD logo files rather than tracing the generated image.

Artwork production record: built-in image generation was used for the concept. The exact generation prompt is embedded in this PDF as an attachment; the manufacturer should work from approved label artwork and a confirmed packaging component.

Promising rationale. Unproven mixture.

Scite-supported paper review, supplemented by manufacturer documents. No retrieved trial tested this VETUD formula.

Darkenyl-associated research / controlled ingredient study

De Tollenaere et al. (2021) studied 44 men over four months with a 1% v/v lotion. Gray-hair density decreased 14% from baseline versus 5% with placebo. The work was supplier-associated and tested a different vehicle; methods and conclusion differ on application frequency. This supports investigation and does not establish VETUD efficacy. [1]

Supplier evidence / Arcolys and SILVERFREE

The Greentech Arcolys summary describes 44 adults, 1% and five months, with a 19% baseline reduction in white-hair density. That is not a 19% placebo-adjusted VETUD benefit. Croda lists 1.5% SILVERFREE and describes clinical studies; complete reports and grade-matched specifications remain required for appraisal. [5,6]

Arey benchmark / supplement plus serum

The 24-week four-arm study reports 173 adults with inconsistent enrollment/completion wording. Its combination improved a scaled score by 13%; mean raw gray-hair counts changed 35.11 to 34.38, while placebo rose 38.29 to 47.81. Serum alone showed little objective change. Brand involvement, one-tailed tests and missed-final-visit exclusions limit interpretation. The study cannot be transferred to this different serum. [2]

ThriveCo benchmark / uncontrolled finished regimen

The 120-day study enrolled 32 people; 29 completed using a multi-ingredient serum and adjunct shampoo. Its 46.28% improvement concerned an ordinal severity score, not 46% fewer gray hairs. Objective lightness change had p=.12. The study does not establish the eight-week percentage claim or isolated Arcolys performance. [3]

Development conclusion: retain one candidate for disciplined testing. Proposed complementary mechanisms do not demonstrate synergy, a guaranteed reversal effect, permanence or superiority. A newly branded package does not change that evidence status.

What the factory should return

Use this revision as the master R&D; request. Return unresolved items explicitly before a manufacturing-release decision.

Requested package: qualified raw-material register; reconciled % w/w master formula and 1 kg quantities; full INCI; manufacturing method with justified limits; pilot batch record; proposed QC specifications; preservation/microbial reports; stability and package results or study plan; safety assessment; measured density/fill calculation; proposed directions and claim substantiation; 50 mL packaging quote, MOQ, cost and lead time.

Open decisions: current active grades and shared process/final-pH window; final dose and use directions; component compatibility; release limits; launch markets and corresponding label requirements. Obtain approval of formula, process, pack and required evidence before production.

References

[1] De Tollenaere, M., Chapuis, E., Auriol, P., Auriol, D., Scandolera, A., & Reynaud, R. (2021). Global repigmentation strategy of grey hair follicles by targeting oxidative stress and stem cells protection. Applied Sciences, 11(4), 1533. Source

[2] Bhardwaj, V., Rodgers, N., Oliveira, C. D., Qin, E., Conrad, A. A., & Conrad, K. A. (2026). Randomized double-blind, placebo-controlled study of a supplement and serum on grey hair. Journal of Drugs in Dermatology, 25(11), 932701. Source

[3] Merja, A. M., Patel, N., Patel, M. N., Patnaik, S., Ahmed, A., & Maulekhi, S. (2023). Safety and efficacy of Arcedin infused anti-gray hair serum in adult human subjects with partially gray hair. Journal of Dermatology & Cosmetology, 7(4), 115-123. Source

[4] Givaudan. Darkenyl, leaflet12.18, technical information p.8. Historical manufacturer document; obtain current grade guidance. Source

[5] Schlegel, P. / Greentech (2021). Reverse hair greying. COSSMA supplier clinical summary. Source

[6] Croda Beauty / Sederma. SILVERFREE product information. Public blend INCI and recommended1.5% addition. Source

[7] Ashland (2024). Microbial protection portfolio. euxyl PE9010 90:10 preservative blend guidance. Source

[8] ISO. ISO11930:2019, preservation efficacy, with applicable amendments; ISO17516:2026, microbial limits. Confirm current applicable requirements. Source

Source limits: Scite retrieved/verified the journal records; publisher text was used where Scite did not return full text. Supplier full confidential dossiers, a VETUD sample and analytical results were not available. Review date: 10 November 2026. Brand artwork is a development concept; it is not evidence of outcome.